

Adaptive Spatio-Temporal Traffic Surveillance

Phase 2 — Model Training & Granular Helmet Compliance

Milestone	Phase 2 of 4 (Days 6–12): Model Training & Helmet Compliance
Author	Harsh Pandhe
Repository	https://github.com/harsh-pandhe/traffic-surveillance-system
Wheel model	YOLOv8-cls selected — 86.7% accuracy (4 classes)
Helmet model	YOLOv8n — mAP@50 0.731 (7 classes)

Executive summary. Phase 2 delivers two trained models on real data. For wheel-count classification, a custom SmallCNN baseline is benchmarked against YOLOv8-cls across four classes (2/3/4/6+ wheelers); YOLOv8-cls is selected at 86.7% accuracy vs 63.9% for the CNN. For helmet compliance, a YOLOv8n detector is fine-tuned on a 7-class rider/helmet dataset, reaching mAP@50 0.731. All metrics are recomputed from the datasets and persisted in results/phase2/.

1. Wheel-Count Classification — CNN vs YOLO

Four classes: 2-Wheeler (bicycle/motorcycle, COCO), 3-Wheeler (auto-rickshaw crops), 4-Wheeler (car), 6+ Wheeler (bus/truck). Balanced ~308–350 train images per class, 96×96 crops, CPU training.

Model	Accuracy	Macro-P	Macro-R	Macro-F1	Latency (ms)
SmallCNN (baseline)	0.639	0.640	0.637	0.627	1.9
YOLOv8-cls (selected)	0.867	0.865	0.867	0.865	3.2

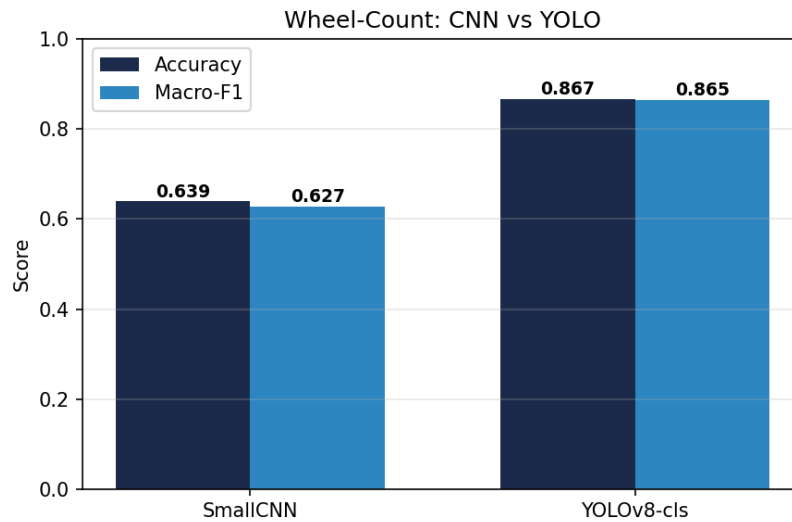


Figure 1. Accuracy and macro-F1: CNN vs YOLO.

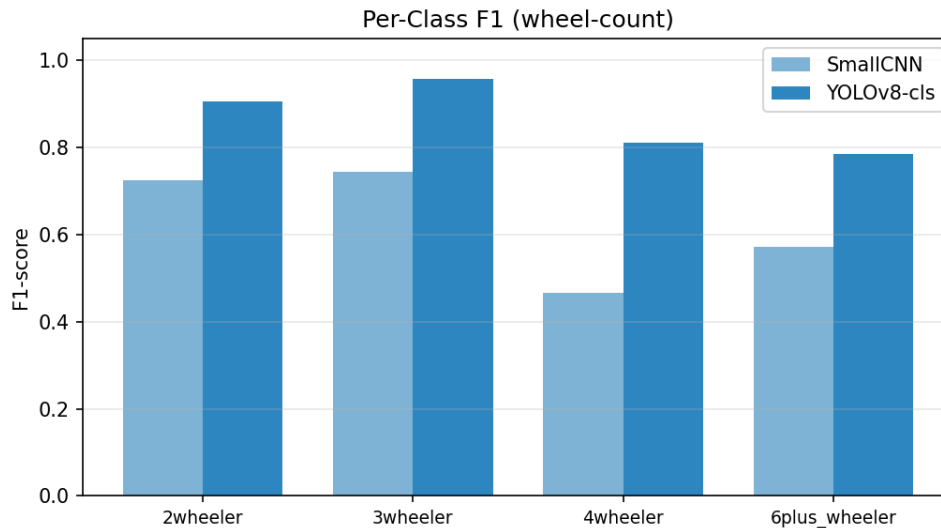


Figure 2. Per-class F1. YOLO wins every class; 3-Wheeler is the strongest (F1 0.96).

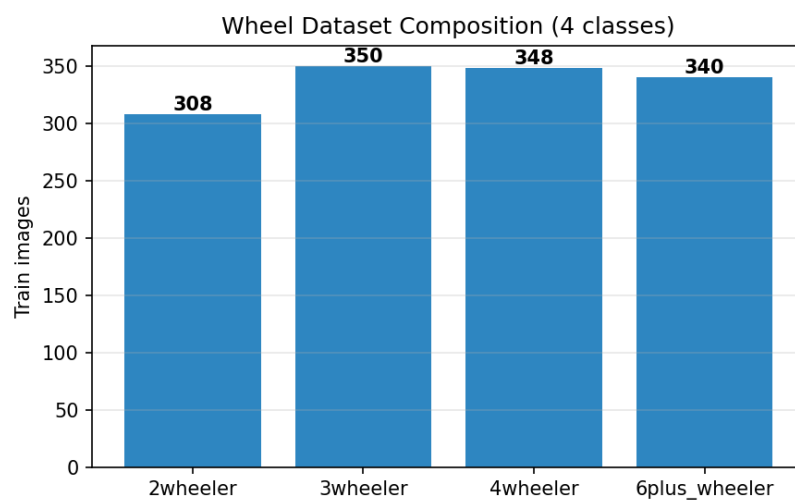


Figure 3. Balanced 4-class wheel dataset.

2. Granular Helmet Compliance — YOLOv8n

7-class rider/helmet detection (driver/passenger × helmet/no-helmet + bike), 368 train / 65 val / 52 test, YOLO format. Fine-tuned YOLOv8n, 25 epochs, 416×416, CPU.

Metric	Value
mAP@50	0.731
mAP@50-95	0.385
Precision	0.741
Recall	0.668

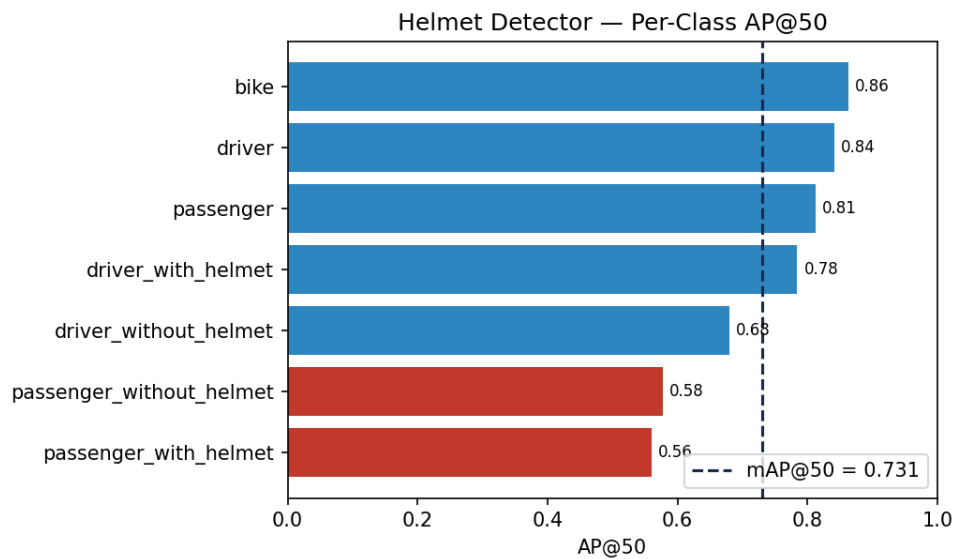


Figure 4. Per-class AP@50 (red = below the mAP line).

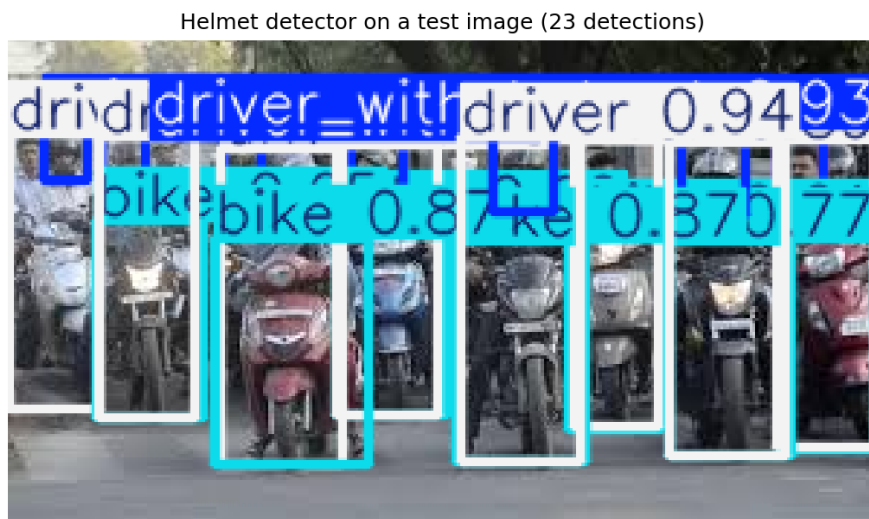


Figure 5. Trained detector on a real test image — riders, bikes, and helmet status.

Passenger-helmet classes are weakest (fewest instances, small objects) — tracked as follow-up issue #8.

3. Reproduce

Command	Purpose
python scripts/build_wheel_dataset.py	Build 4-class wheel crops
python scripts/train_wheel_classifier.py	Train + benchmark CNN vs YOLO
python scripts/train_helmet_detector.py --data ...	Train + eval helmet detector
python scripts/make_phase2_assets.py	Regenerate this report's figures

4. Status & Follow-ups

Deliverable	Status
Baseline CNN trained	Complete
YOLO model trained (wheel + helmet)	Complete
P/R/mAP comparison + selected model	Complete
3-Wheeler class (auto-rickshaw)	Complete (issue closed)
Granular helmet detector (7-class mAP)	Complete
Low AP passenger-helmet classes	Open (issue #8)

Conclusion. Phase 2 is complete: both models trained and benchmarked on real data, all four wheel classes active, and the granular helmet detector reporting per-class AP. Every figure is reproducible from the committed code.